Methods description — MVA Hackathon 2026, Track 1 (Variant Prediction)

Please submit your answers with your Track 1 submission. All questions (with the exception of Question 1, in red) are voluntary; however, submissions without a methods description may be judged less favorably on the Innovation and Scalability criteria, since the panel can't assess what it can't see.

Fill out this form once per model/approach submitted.

	Answers
Team name	Lobo_MX
Model number (fill out this form once per	4
Please describe your model/approach in detail.	We selected the genes from gene expression omnibus information, get it from here: Public resources included GEO Series GSE197267 describing biallelic loss-of-function mutations in MAD1L1 associated with an MVA-like syndrome, GEO Series GSE134780 describing the role of BUB1B, CEP57, and TRIP13 in Mosaic Variegated Aneuploidy syndrome. then we got the coordinates from NCBI, e.g. https://www.ncbi.nlm.nih.gov/gene/?term=BUB1B NC_000015.10:40161069-40221123 Next scanning the VCF filer we got: Found 14 alternative variants in BUB1B Found 4 alternative variants in CEP57 Found 3 alternative variants in TRIP13 Found 889 alternative variants in MAD1L1 We ignored MAD1L1, the number 889 means noise. Next using VEP online interface (https://jun2026.archive.ensembl.org/Tools/VEP) Then we filtered the results by IMPACT e.g. BUB1B 15 40168264 40168264 G/A 15 40180642 40180643 CT/C 15 40181093 40181094 AT/A 15 40182809 40182809 A/C 15 40186692 40186692 G/A 15 40187404 40187404 A/G 15 40192892 40192892 C/T 15 40193435 40193435 T/C 15 40198239 40198239 G/GT 15 40209701 40209701 T/G 15 40216470 40216470 A/G 15 40216568 40216569 TA/T 15 40216570 40216571 TA/T 15 40220612 40220612 T/G
(required) If commercially-available Generative AI was used, record the provider, the plan or tier, and the relevant setting in your methods description. A line is enough, for example, "Anthropic API,	Microsoft Copilot (GPT-5-based M365 Copilot), commercial service. Used for code review, method documentation, and preparation of the submission report. No patient-identifying data were provided beyond the challenge materials.

Please state whether the submission file is the automated output of your computational approach	The submitted candidate list is the automated output of the computational pipeline, followed by limited downstream manual interpretation used only to prioritize the final candidate genes and prepare the written report.
Please describe any downstream manual review in detail.	Manual review focused on candidate MVA genes (BUB1B, CEP57, and TRIP13). Variants extracted from the VCF were annotated with Ensembl VEP and reviewed for predicted consequence, impact level, pathogenicity evidence, and population frequency. Transcript-level annotations were collapsed to unique genomic variants, and coding variants were prioritized over intronic or modifier variants. This process identified a pathogenic BUB1B stop-gained variant (chr15:40209701 T>G) and a damaging BUB1B missense variant (chr15:40220612 T>G) as the strongest compound
Please state whether your approach only used publicly available data (preferred) or if proprietary data was also used. If proprietary data were used, where possible, please also	This approach used only publicly available data sources in addition to the challenge-provided WGS dataset.
Please describe any public data used in detail (e.g., reference panels, population databases, literature).	Public data sources included Gene Expression Omnibus (GEO) datasets GSE197267 and GSE134780, which were used to identify genes previously associated with Mosaic Variegated Aneuploidy (MVA) syndrome and MVA-like phenotypes, specifically BUB1B, CEP57, TRIP13, and MAD1L1. Genomic coordinates for the candidate genes were obtained from the NCBI Gene database. Functional annotation and variant consequence prediction were performed using the Ensembl Variant Effect Predictor (VEP), incorporating reference data from ClinVar for clinical significance, gnomAD and 1000 Genomes for population allele frequencies, MANE transcripts for transcript prioritization, and in silico pathogenicity prediction tools including SIFT, PolyPhen, ClinPred, CADD, and SpliceAI.
Please describe any proprietary data used in detail.	No proprietary datasets, databases, annotation tools, or commercial genomic resources were used.
Is your approach able to output proposed pairs of compound heterozygous candidate variants, or only single	The approach supports both single candidate variants and proposed compound heterozygous variant pairs within the same candidate gene. The workflow was specifically designed to identify recessive and potential compound heterozygous events consistent with published MVA disease
How did you handle secondary or incidental findings, if any? (See the finding_type column in your submission.)	Secondary findings were retained in the output and labeled using the finding_type = secondary field. Findings were downgraded to secondary status when confidence was lower than primary candidates, such as variant pairs located in very close genomic proximity that may represent in-cis events rather than true compound heterozygous configurations. Secondary findings were included to preserve potentially relevant
Please provide an estimate of run time and cost for your approach (if possible).	Runtime: approximately 1 to 5 minutes depending on hardware. Hardware requirements: standard desktop or laptop CPU. GPU requirements: none. Computational cost: negligible. Software cost: open-source Python environment.

Provide a method abstract (up to 500 words). Please include strengths and limitations of your method as well as methodological detail.

Mosaic Variegated Aneuploidy (MVA) syndrome is a rare chromosomal instability disorder that has been associated with pathogenic variants in several mitotic checkpoint genes, including BUB1B, CEP57, TRIP13, and more recently MAD1L1. To prioritize candidate disease-causing variants, a targeted gene-driven strategy was performed using publicly available Gene Expression Omnibus (GEO) resources. GEO Series GSE197267 reported biallelic loss-of-function mutations in MAD1L1 causing an MVA-like syndrome, while GEO Series GSE134780 highlighted the established role of BUB1B, CEP57, and TRIP13 in classical MVA syndrome. Genomic coordinates for these candidate genes were subsequently retrieved from NCBI Gene and used to interrogate the whole-genome sequencing variant call file.

Variant extraction from the VCF identified 14 alternative variants within BUB1B, 4 variants within CEP57, 3 variants within TRIP13, and 889 variants within MAD1L1. Because the extremely large number of observed MAD1L1 variants suggested substantial background variation and reduced interpretability, subsequent analyses focused on the three canonical MVA-associated genes: BUB1B, CEP57, and TRIP13. Functional characterization of all candidate variants was performed using the Ensembl Variant Effect Predictor (VEP).

Following annotation, variants were prioritized according to predicted functional impact. Most variants identified in BUB1B, CEP57, and TRIP13 were classified as intronic, modifier, upstream, or downstream variants and therefore provided limited evidence for pathogenicity. However, one BUB1B variant, chr15:40209701 T>G, was consistently annotated across multiple transcripts as a HIGH-impact stop-gained variant, including annotations of stop_gained and stop_gained, NMD_transcript_variant. This variant represents a premature truncation event in a gene that is a well-established cause of MVA syndrome. A second BUB1B variant, chr15:40220612 T>G, was consistently annotated as a MODERATE-impact missense variant. In corresponding transcript annotations, this variant was predicted to be deleterious by SIFT, probably damaging by PolyPhen, and received a high ClinPred score, supporting a functional effect on the encoded protein.

Among CEP57 variants, the strongest finding was chr11:95795499 C>CT, which was annotated as a splice_polypyrimidine_tract_variant and intron_variant with LOW predicted impact. While potentially relevant because of its location in a known MVA gene, the evidence supporting pathogenicity was considerably weaker than that observed for BUB1B. Similarly, the analyzed TRIP13 variants were predominantly intronic modifier variants and lacked damaging coding consequences.

Taken together, the evidence strongly supports a compound heterozygous BUB1B model, consisting of the HIGH-impact stop-gained variant chr15:40209701 T>G and the MODERATE-impact damaging missense variant chr15:40220612 T>G. These variants represent the most plausible disease-causing candidate pair and were therefore prioritized as the primary finding for submission.